In Person Public Health Training

Interpreting Hepatitis B Serological Laboratory Results: Handout

Justinian Wurtzel, MSc MPH

DrPH Student, Epidemiology

CPH 705: Public Health Teaching and Training

Dr. Analisa McMillan, PhD

November 2026

Viral Hepatitis B, Quick Reference

Purpose

This document was designed for providers and other hepatitis B related staff to use as a quick reference to review testing, clinical information, and other relevant viral hepatitis B information to support their patients and decision making.

Testing

Term	Definition
Screening	Testing people with no symptoms (asymptomatic) to find possible infection or immunity (one time CDC recommended adult triple panel test)
Testing (Diagnostic)	Testing people with symptoms (symptomatic), a plausible exposure, or prior positive result to confirm infection
Antigen	A non-self molecule that human immune systems use to identify pathogens (HBsAg, HBeAg)
Antibody	A protein produced by the human immune system to attack pathogens (Anti-HBs, Anti-HBc, Anti-HBe)
Qualitative	Laboratory result reported as a positive/negative or reactive/non-reactive
Quantitative	Laboratory result reported as a numeric value (anti-HBs>=10 mIU/mL, 2,000 HBV DNA IU/mL)
Detectable	A qualitative measure of disease antigen being present at the limit of a laboratory test or over
Undetectable	A qualitative measure of disease antigen being undetected by the laboratory test

Infection and Immunity

Term	Definition
Acute	Previously not infected with HBV, a newly acquired infection within the first six months of exposure
Chronic	HBsAg and or HBV DNA persisting over six months after acute status
Spontaneous Resolution	Immune system clears HBV without supportive treatment and develops protective antibodies
Immune due to natural infection	Protection after a previous infection, resolved due to supportive therapy or spontaneous resolution
Immune due to vaccination	Protection after HBV vaccine series developing protective antibodies anti-HBs
Reactivation	A flare up of someone with a chronic or resolved infection previously suppressed and not infectious, often seen in older patients or those with immunocompromised status

Laboratory Tests

Term	Definition
HBsAg	HBV surface antigen protein on the surface of the virus
Anti-HBs	Antibody to surface antigen, made by human immune system
Total IgM/IgG Anti-HBc	All immunoglobulin (antibodies) to the viral antigen core
IgM Anti-HBc	Core antibody mark of acute case within first six months of exposure
HBeAg	HBV e-antigen protein an indicator of active viral replication
Anti-HBe (HBeAb)	Antibody to the e-antigen, made by the human immune system
HBV DNA	The genetic material of HBV, used quantitatively for a viral load
Triple Panel	HBsAg, Anti-HBs, Total IgM/IgG Anti-HBc

Clinical Information

Term	Definition
Jaundice	Yellowing of the skin or eyes, can also be detected in urine and stool
Bilirubin	A breakdown of red blood cells processed by liver, Bilirubin>=3.0mg/dL for acute criteria
ALT, Alanine Aminotransferase	An enzyme found primarily in the liver, released when liver cells are damaged, ALT> 200 IU/L
AST, Aspartate Aminotransferase	An enzyme found around the body and in the liver, identified in liver damage but not diagnostic
HBIG, Hepatitis B Immunoglobulin	External hepatitis B immunoglobulin (antibody) given after exposure typically used for infants born to hepatitis B positive mother and also to people exposed via needlestick or sexual contacts

Surveillance Terms and Priority Populations

Term	Definition
Reportable disease	A disease that by law is required to be reported to the state health department, any positive hepatitis B laboratory results are required to be sent to Iowa HHS
Confirmed case	Meets the confirmatory laboratory criteria (and clinical criteria for acute case) from the 2024 CDC case definitions[1]
Probable case	Meets presumptive laboratory criteria (acute with clinical criteria)
Not a case	Does not meet confirmatory or presumptive laboratory criteria
PWID	People who inject drugs, person-first language for a priority population with higher exposure risk to hepatitis B
MSM	Men who have sex with men, a priority population with higher exposure risk to hepatitis B
Country of Origin	Identifies if a patient is US-born or if from a country with a higher prevalence of hepatitis B, helps determine likelihood of chronic status

CDC 2024 Case Classification[1]

Classification	Laboratory Evidence	Clinical Criteria
Acute, Confirmed Tier 1	Positive HBsAg, HBeAg, or HBV DNA with IgM Anti-HBc or seroconversion within 12 months of negative HBsAg	No
Acute, Confirmed Tier 2	Positive HBsAg or HBV DNA with IgM Anti-HBc not	Yes
Acute, Probable	Positive IgM Anti-HBc with HBsAg, HBeAg and HBV DNA negative or not done	Yes
Chronic, Confirmed	Positive HBsAg/HBeAg twice six months apart or HBsAg/HBeAg plus Total IgM/IgG Anti-HBc or HBsAg plus HBeAg or HBV DNA	No
Chronic, Probable	HBsAg or HBeAg with negative IgM Anti-HBc or Not done	No

Interpretation of Laboratory Results[2]

HBsAg	Total Anti-HBc	IgM Anti-HBc	Anti-HBs	Interpretation
-	-	-	-	Never infected, susceptible, vaccinate
+	-	-	-	Early acute or false positive up to 18 days after vaccination
+	+	+	-	Acute infection
-	+	+	-	Acute infection, resolving
-	+	-	+	Recovered from past infection, immune
+	+	-	-	Chronic infection
-	+	-	-	Isolated core, past infection, low level chronic, or false positive
-	-	-	+	Immune from vaccination (Anti-HBs>=10mIU/mL)

References

1. Hepatitis B, acute and chronic 2024 Case Definition | CDC. May 30, 2024. Accessed November 6, 2026. https://ndc.services.cdc.gov/case-definitions/hepatitis-b-acute-and-chronic-2024/

2. Table 5 - STI Treatment Guidelines. July 15, 2021. Accessed November 6, 2026. https://www.cdc.gov/std/treatment-guidelines/table-5.htm